AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MST129
Applied Calculus

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 20/Apr./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 90 minutes**

Instructions:

- Total Marks: **60**
- This exam consists of **two** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

Q–1: [5×3 marks] choose the correct answer:

1.1: The domain of the function f(x)= $\frac{x}{\sqrt{x+1}}$ is:

a) [-1, 1]
b) $(-1,1)$
c) $(-1,\infty)$
d) $(-\infty,1)$
e) None of the above

1.2: The second Derivative of $f(x) = (5x + 4)^3$ at x= -1 is:

a) -75
b) 75
c) 15
d) -15
e) None of the above

1.3: The tangent line to the curve of f(x)= $4x^2 - 8x$ *is* horizontal at $x =$

a) 1
b) - 1
c) 3
d) 6
e) None of the above

1.4 : The slope of the tangent line to f(x)= $\sqrt{x}$ at x= 4 is:

a) $-\frac{1}{4}$
b) $\frac{1}{4}$
c) 4
d) -4
e) None of the above

1.5: The (instantaneous) rate of change of f(x)= $x^2 - 2x + 3$, *when* $x = 5$ is:

a) 2
b) 15
c) 0
d) 8
e) None of the above

Arab Open University
القبول والتسجيل

CamScanner

Each question is worth 15 marks. Answer the following questions:

Q-2:

a)[6 marks] Find the points of intersection of the curve $y= -x^2 +x+1$ and $y=x -5$.

Let) $f(x) = 3 + \frac{1}{(x+1)^3}$, and $g(x) = 4x^3 - 2x^2 + x + 1$

i) [3+3 marks] Find the derivative of $f(x)$ and $g(x)$.

ii) [3 marks] Find $(f \circ g)(x)$.

Q–3:

a) [8 marks]Find the slope of the tangent line to curve $y= x^3 - 4x^2 +6x +1$ at $x=2$, then write the equation of this tangent line.

b) [2+2+3 marks] The position at time $t \geq 0$ of a particle moving in a straight line is given by $S = t^4 - 4t^3 + 4t^2$,

i) Find the particle's velocity and acceleration at time t.
ii) When does the particle return to its initial position?

Q–4: Let $f(x) = -2x^4 + 4x^2$

a) [4 marks] Find $f'(x)$ and $f''(x)$.

b) [11 marks] Find the intervals on which f is increasing or decreasing and find the local maximum and minimum values of $f(x)$, if exist, by using first derivative test.

End of questions

Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt